# Notional Institute of Technology, Kurukshetra

**BTech(3rd Sem), S-1**

**Subject & code: Software Engineering (CSPC- 304)** **D.O.E: 20/02/2025**

**Time: 50 mts** **Marks: 15**

Attempt all the questions. All question carry equal marks.

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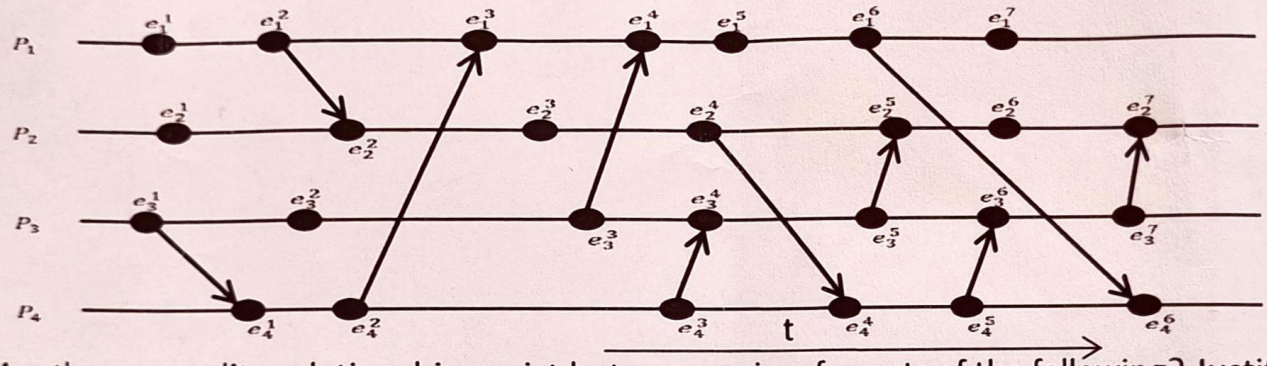
|---|---|---|
| Q1. | Distinguish between synchronous and asynchronous distributed systems. | (5) |
| Q2. | Consider the space-time diagram given below shown 4 processes.<br>  <br><br>Are there causality relationships exist between pairs of events of the following? Justify your answer.<br>(i) $e_1^2$ and $e_4^4$ (ii) $e_3^2$ and $e_2^5$ (iii) $e_2^1$ and $e_4^3$ (iv) $e_3^4$ and $e_2^7$ | (5) |
| Q3. | Consider the above space-time diagram, where processes start its execution with vector time stamps $P_1(0, 0, 0, 0)$, $P_2(0, 0, 0, 0)$, $P_3(0, 0, 0, 0)$ and $P_4(0, 0, 0, 0)$. Find the vector clock values for all the events. | 5 |